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Rock Piles as Kōura Habitat

Reinforcing kōura resilience using rock piles to enhance freshwater crayfish habitat
in an Aotearoa New Zealand lake

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Placeholder — abstract to be written.

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24 Abstract

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26 Introduction

27 Freshwater biodiversity is declining faster than any other biome, with one quarter of freshwater
28 fauna now threatened with extinction (Sayer *et al.*, 2025). Among the most endangered are
29 freshwater crayfish, with 32% of species assessed as threatened globally due to habitat loss,
30 invasive species, and water quality decline (Richman *et al.*, 2015). As keystone species and
31 ecosystem engineers, crayfish play crucial roles in nutrient cycling, bioturbation, and foodweb
1 dynamics, and their loss can have cascading consequences for freshwater ecosystems (Momot,
2 1995; Collier, Parkyn & Rabeni, 1997; Parkyn, Collier & Hicks, 2001).

3 In Aotearoa New Zealand (hereafter Aotearoa), the endemic freshwater crayfish kōura
4 (*Paranephrops planifrons* White, 1842) is a taonga (culturally treasured) species of significant
5 ecological and cultural importance to Māori, particularly for Te Arawa iwi of the Rotorua lakes
6 region in the North Island of Aotearoa (Kusabs & Quinn, 2009). Kōura support customary
7 fisheries using cultural practices such as tau kōura, yet populations in Lakes Rotoiti and
8 Rotorua have declined by up to 96 % over the past two decades, with consequent loss of
9 these customary practices (Kusabs *et al.*, 2015a; Kusabs *et al.*, 2026). The primary driver of
10 decline is predation by the invasive brown bullhead catfish (*Ameiurus nebulosus* Lesueur, 1819,
11 hereafter catfish), confirmed in these lakes in 2016 and 2018 (Francis, 2019; Grayling, 2020).
12 Catfish are native to eastern North America where they mostly live in muddy slow-moving
13 water bodies and can survive and spawn in a wide temperature range, tolerate poor water
14 quality, low oxygen concentration and have a broad diet (Scott & Crossman, 1973), making
15 them a highly adaptable/tolerant invader. Catfish were introduced in Aotearoa in 1877 as
16 sport fish (McDowall, 1994) with their distribution continuously spreading over the upper
17 North Island such as the Waikato River and in 1985 they were detected in Lake Taupō (Barnes
18 & Hicks, 2003). In Aotearoa catfish are a major predator on kōura, as 64 % of large catfish
19 (<250 mm fork length) had kōura in their stomach contents (Barnes & Hicks, 2003). In
20 combination with predation by catfish kōura are further compressed by multiple interacting
21 stressors that progressively restrict their available habitat. Kōura are naturally found in
22 complex rocky habitats (Jowett, Parkyn & Richardson, 2008; Kusabs, Quinn & Hamilton,
23 2015b) or in deeper parts of the lake to avoid predation during the day, migrating to the

productive littoral zone at night to forage (Deveich, 1979). However, thermal stratification creates anoxic conditions in deeper waters, displacing kōura into the shallow littoral zone (Kusabs *et al.*, 2026), precisely where catfish predation pressure is greatest, as catfish rarely swim beyond 17 m depth (Dedual, 2002). This compression is further intensified by dense invasive macrophyte beds obstructing kōura migration in the littoral zone (Kusabs & Quinn, 2009), which also intensify oxygen depletion in deeper waters by die-off events (Vincent, Gibbs & Dryden, 1984; Hamilton, McBride & Uraoka, 2005). The importance of deep refuge is shown by Lake Taupō, where kōura persist at a depth even in the presence of catfish because the lake remains oxygenated below the thermocline (Verburg & Albert, 2019). In Lake Rotoiti this deep refuge becomes anoxic and unavailable for kōura (Kusabs *et al.*, 2026). Next to direct predation catfish exposure triggers anti-predator responses in kōura including increased refuge use (Raven *et al.*, 2026a), reducing time available for foraging and potentially suppressing growth and reproduction (Preisser, Bolnick & Benard, 2005; Sheriff *et al.*, 2020). Climate change is expected to strengthen thermal stratification, expand anoxic zones (Woolway *et al.*, 2021), and increase the range of suitable catfish habitat in Aotearoa, further reducing kōura persistence (Lee *et al.*, 2025).

The compression of kōura into the littoral zone makes the quality of nearshore habitats increasingly important for kōura populations. In stream environments, kōura are positively associated with structurally complex habitats including bank undercuts, woody debris, leaf litter, and coarse substrates (Usio & Townsend, 2000; Parkyn & Collier, 2004; Jowett *et al.*, 2008; Parkyn, Meleason & Davies-Colley, 2009). In lake environments benthic substrate characteristics are a stronger predictor of kōura distribution than nutrient enrichment or predator abundance (Kusabs *et al.*, 2015b), and kōura are positively associated with coarse substrates and riparian vegetation (Raven *et al.*, 2026b). Coarse rocky substrate provides refugia from predators and shelter during moulting (Streissl & Hödl, 2002). Experimental studies confirm that shelter availability directly reduces catfish predation on crayfish (Adams, 2007; Zhao *et al.*, 2024), and kōura increase refuge use in the presence of catfish compared to native predators (Raven *et al.*, 2026a), emphasising the importance of shelter availability for kōura populations. Shelter availability is likely a limiting factor for kōura in Lake Rotoiti, where the majority of the shoreline is sandy with dense beds of introduced macrophytes and little natural rocky structure. Adding rocky substrate to soft-bottomed lake areas has been shown to significantly increase crayfish abundance (Johnsen & Taugbøl, 2008), and increasing structural complexity in streams improves crayfish restocking success (Huolila, Marjomäki & Laukkanen, 1997). Deploying habitat enhancement structures that add structurally complexity in the littoral zone could potentially be a practical way to restore kōura populations in Lake

25 Rotoiti.

26 In this study habitat enhancement structures in the shape of stone piles were deployed along
27 the shoreline of Lake Rotoiti to investigate the ability of the structures in providing additional
28 habitat for kōura to live and hide from predators. Kōura presence, abundance, and biomass
29 were monitored at stone pile, control, and reference sites over 18 months. We predicted that
1 kōura occupancy, abundance and biomass would increase at stone pile sites relative to control
2 sites after construction, and that catfish would not change. We further predicted that kōura
3 abundance would keep increasing progressively after construction, at least within the first 18
4 months, and that catfish would instead show seasonal patterns of shoreline use rather than a
5 construction-driven trend (Barnes, 1996; Dedual, 2002). Finally, we predicted that stone pile
6 sites would progressively converge toward kōura densities observed at natural rocky reference
7 sites, and that catfish would not show this pattern.

8 **Materials and Methods**

9 **Study location and construction**

10 Lake Rotoiti (-38.03, 176.42) is a warm monomictic mesotrophic lake located in the Rotorua Te
11 Arawa Lakes region of the North Island of Aotearoa (34 km², maximum depth 128 m). The lake
12 stratifies for approximately nine months a year. Since 2008, nutrient-rich inflows from Lake
13 Rotorua have been diverted toward the Kaituna River via the Ōhau Channel diversion wall to
14 improve water quality in Lake Rotoiti (Hamilton *et al.*, 2005; Prentice & Özkundakci, 2024).
15 Lake level is regulated by a weir at the outlet of the Kaituna River, providing a stable and
16 controlled water level throughout the year. Catfish are present throughout the lake but occur
17 in particularly high concentrations in Te Weta Bay on the North West side of the lake (Francis,
18 2019). Five stone pile sites were selected alongside the lake shoreline based on accessibility and
19 land owner approval (Figure 1). At each stone pile site, a paired control site was established
20 50 m along the shoreline in the same habitat type. Five naturally occurring rocky shoreline
21 locations were selected as reference sites, each with two monitoring points 50 m apart. In total
22 20 sites of 150 m² (15 * 10 m) were monitored seasonally over 18 months. Baseline monitoring
23 was conducted in February 2025 prior to stone pile deployment, which occurred the following
24 week. Stone piles were constructed from locally sourced quarry rock (Manawahe Stone and
25 Quarry, ~25 km from the lake) with a size range of 50 - 400 mm. At each of the five stone pile
26 sites, three stone piles were arranged horizontally along the shoreline by hand from the shore.

27 Stone pile design was informed by mesocosm experiments demonstrating kōura preference for
28 enclosed stone structures (Raven *et al.*, in review). Stone piles were positioned approximately
29 4.3 m from shore at a mean depth of 0.49 m and with a mean area of 4.1 m².

30 **Monitoring**

31 Monitoring consisted of physical, chemical, and biological parameters recorded at each site
32 during each monitoring round. Static site characteristics including substrate composition, slope
1 to depth contour, riparian vegetation, overhanging tree cover and wood cover were visually
2 estimated at baseline monitoring only, as these were not expected to change substantially over
3 the monitoring period. Substrate composition was visually estimated using an underwater
4 viewing chamber, categorising bedrock, boulders (>256 mm), cobble (64–256 mm), gravel (2–64
5 mm), sand (0.063–2 mm), mud (<0.063 mm), and organic matter. A substrate index was
6 calculated from these estimates (Jowett & Richardson, 1990; Jowett *et al.*, 2008):

$$\text{Substrate index} = 0.08 \times \text{bedrock} + 0.07 \times \text{boulders} + 0.06 \times \text{cobble} + 0.04 \times \text{gravel} + 0.03 \times \text{sand} + 0.02 \times \text{organic matter}$$

7 Coarser substrates were assigned higher weights as they provide greater interstitial pore space
8 and stable structures used by kōura as refuge (Usio & Townsend, 2000; Nyström *et al.*, 2006).
9 After stone pile construction, measurements of the size and dimensions of each structure were
10 taken. The post-construction substrate index for stone pile sites was recalculated using each
11 site's combined stone-pile footprint area, expressed as a percentage of the 150 m² site area
12 and treated as 80 % cobble and 20 % boulders displacing an equivalent area of sand, to reflect
13 their contribution to substrate complexity. Slope was assessed as the horizontal distance from
14 the shoreline to the 20 m depth contour, derived from bathymetric maps and measured in a
15 projected coordinate system (NZTM2000; EPSG:2193).

16 Seasonal variable parameters were recorded at every monitoring round, and included aquatic
17 vegetation cover (emergent and submerged macrophytes) and chemical parameters included
18 water temperature (°C), dissolved oxygen (mg L⁻¹), and conductivity (µS cm⁻¹) were measured
19 using a YSI ProSolo meter (Pro2030, YSI Inc., Yellow Springs, OH, USA). pH was measured
20 using a handheld pH meter (EC-PCTestr35, Eutech Instruments).

21 Kōura and fish were every monitoring round sampled using two single-winged, double-throated
22 fyke nets (600 mm high D-shaped entrance hoop) and two collapsible bait traps (480 × 250 mm;
23 two 50 mm entrance openings) baited with catfish flesh (MacNeil *et al.*, 2025). Fyke nets were

placed perpendicular to the shoreline outside of the stone pile structures or at 10 m intervals, with bait traps positioned between them. All equipment was deployed simultaneously at each site for a 24-h period. Captured kōura were measured for orbital carapace length (OCL) using a digital calliper (0.01 mm precision; 150 mm) and for wet mass using a digital scale (± 1 g; 5 kg capacity, Anko Brand, Kmart Group, Mulgrave, VIC, Australia). Catfish and other fish species, including eels (*Anguilla* spp.), goldfish (*Carassius auratus* (Linnaeus, 1758)), common smelt (*Retropinna retropinna* (Richardson, 1848)), kōaro (*Galaxias brevipinnis* Günther, 1866), rainbow trout (*Oncorhynchus mykiss* Walbaum, 1792), common bully (*Gobiomorphus cotidianus* McDowall & Fulton, 1978), and mosquitofish (*Gambusia affinis* (S. F. Baird & Girard, 1853)), were recorded for abundance and measured for total length, and weighed for wet mass. Catch per unit effort (CPUE) and biomass per unit effort (BPUE) were calculated for all kōura and fish based on the number of nets and traps used at each site for each monitoring round (Kusabs *et al.*, 2015b). Missing kōura weights due to failure of scale ($n = 36$) were estimated using a log $-\log$ length–weight regression fitted to individuals with both measurements ($n = 757$), following standard approaches for weight length relationships in aquatic organisms Figure 5. Predicted values were back-transformed with a lognormal bias correction factor ($10^{\{0.5 \cdot \hat{\sigma}^2\}}$) (Quinn & Keough, 2002).

$$\log_{10}(\text{Weight [g]}) = -2.67 + 2.683 \times \log_{10}(\text{OCL length [mm]})$$

12 Data analysis

Data analysis was divided into three sections, the first section assessed baseline equivalence between stone pile, control, and reference sites before construction, the second section assessed the impact of stone pile construction and after-construction trajectory on kōura and catfish occupancy, abundance, and biomass, and the third section assessed whether stone pile sites converged toward reference sites and if reference sites remained stable over time. All analysis was conducted in R (version 4.5.2; R Core Team (2025)) using the glmmTMB and emmeans packages (Brooks *et al.*, 2017; Lenth & Piaskowski, 2017).

Site variation before stone pile construction was assessed using paired Wilcoxon signed-rank tests to compare stone pile sites with their paired control sites, and unpaired Wilcoxon rank-sum tests to compare stone pile sites with reference sites. This was done for response variables (water quality, substrate, CPUE, BPUE) to confirm that the stone pile and control sites were equivalent before construction, and to characterise the difference between stone pile and

reference sites. To further assess the baseline equivalence generalised linear models (GLMs) were fitted testing the CPUE and BPUE of kōura and catfish against site type (stone pile/control) and tested each of the two reference sites (reference 1/reference 2) with a Tweedie family and log link. The covariate temperature was included in the catfish abundance and biomass models testing the reference sites to allow model convergence.

The second section of the analysis tested construction impact relative to control sites using a before-after, control-impact (BACI) design ([Underwood, 1994](#)) and tested after-construction trajectory of kōura and catfish populations. The BACI design tested using generalised linear mixed effects models (GLMMs) the effect of stone pile construction on kōura and catfish occupancy, abundance, and biomass by comparing one pre-construction monitoring round with six post-construction monitoring rounds. The models tested period (before/after) and site type (stone pile/control) as an interaction term, with monitoring round as a covariate and site nested within pair as a random effect. Occupancy was tested with a Binomial family and abundance and biomass using a Tweedie family with log link. The after-construction trajectory models were tested using GLMMs with site type (stone pile/control) and monitoring round as interaction terms and site number as random effect using similar families as the BACI models.

The third section of the analysis assessed whether stone pile sites converged toward reference sites and if reference sites remained stable over time. Kōura and catfish abundance, and biomass were tested in GLMMs fitted with site type (stone pile/reference) and monitoring round as interaction terms with site number as random effect using a Tweedie family with log link. The catfish abundance model also included temperature as a covariate. The reference site stability was tested in GLMMs fitted with site type (reference 1/reference 2) and monitoring round as interaction terms with site nested within pair as a random effect also using a Tweedie family with log link. Temperature was included in the catfish abundance and season was included in the kōura abundance as covariates.

Multiple-testing correction using Benjamini–Hochberg correction was applied within each research sub-question across the set of models tested. Robustness of the models was further assessed via sensitivity analyses and all models were tested for uniform distribution, dispersion, and outlier rate using the DHARMA package ([Hartig, 2016](#)). Covariate selection for each model was preformed by comparing Corrected Akaike Information Criterion (AICc) of the base model against models with one additional covariate (temperature, season, or substrate index) added at a time. A covariate was retained only when it reduced AICc by more than 2 units relative to the base model ([Advanced issues and deeper insights, 2002](#)). Temperature was

11 retained in in catfish models where it substantially improved model fit, consistent with catfish
12 being associated with warmer water temperatures ([Scott & Crossman, 1973](#)). For the two
13 reference-site catfish models specifically, the base model failed to converge and temperature
14 was required for numerical stability. Season was retained in the kōura abundance model for
15 reference stability, reflecting the known seasonal variation in kōura abundance ([Kusabs *et*
16 *al.*, 2015b](#)). Season could not be evaluated for the BACI models as it is collinear with the
17 before/after period contrast.

18 **Results**

19 Figure [2](#)

20 Figure [4](#)

21 *Baseline equivalence*

22 *Construction impact relative to control sites*

23 *Recovery toward reference conditions*

24 Figure [3](#)

1 **Discussion**

2 *Opening*

3 **Baseline and site context***

4 **Construction impact***

5 **Convergence and recovery***

6 **Implications and future directions***

7 Most kōura were found along the edges of the stone piles there for recommended future stone
8 piles should be designed to create the most edges.

9 Stone piles were constructed at the shore line to shallow to reach macrophyte beds which
10 are located further offshore. Future stone pile construction could be extended further offshore
11 to reach the macrophyte beds and test if stone piles could suppress macrophyte growth and
12 create pathways for kōura to move between the deeper parts of the lake and the shore.

1 **Conclusion**

1 **Acknowledgements**

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2 and study the treasured kōura. We are grateful to Soweeta Fort-D'ath and William Anaru
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13 **Authors' contributions**

14 Study conception and design were undertaken by O.V.R., I.A.K.K., R.H., F.J.B. and D.Ö.
15 Fieldwork was executed by O.V.R. Data analysis was performed by O.V.R. I.A.K.K obtained
16 council and iwi consent and provided local ecological knowledge. Manuscript drafting was led
17 by O.V.R. All authors contributed to manuscript revision, approved the results, and the final
18 manuscript.

19 Use of Artificial Intelligence

20 Use of Artificial Intelligence: Claude (Anthropic) was used to assist with manuscript editing and
21 language refinement. All content was critically reviewed and the authors take full responsibility
22 for the final manuscript.

23 Declarations

24 The corresponding author has declared that none of the authors has any competing interests.

25 Data availability statement

1 The code and derived data supporting the findings of this study are openly available on GitHub
2 (https://github.com/OlivierRaven/Stone_Piles_Rotoiti) and a rendered version of the analysis
3 is hosted at https://oliverraven.github.io/Stone_Piles_Rotoiti/. The primary raw dataset is
4 archived on Zenodo (<https://doi.org/.....>). Peer review documents, including the cover letter
5 and reviewer responses, are available in the manuscript repository in the interest of open and
6 transparent science.

7 Tables

Table 1: Baseline (pre-construction) comparability of stone pile, control, and reference sites. Values are mean $\pm$ SD; p-values test Stone pile vs Control (paired Wilcoxon signed-rank test, pairs matched by pair_id, supporting the BACI design) and Stone pile vs Reference (unpaired Wilcoxon rank-sum test, documenting the starting gap relative to natural conditions, motivating the convergence analysis). With only 5 paired sites, these are low-power checks — a non-significant result indicates no obvious gross imbalance, not formal proof of equivalence.

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Kōura CPUE	0.2 $\pm$ 0.27	0.15 $\pm$ 0.22	2.13 $\pm$ 2.32	1.000	0.205
Catfish CPUE	0.05 $\pm$ 0.11	0.1 $\pm$ 0.22	0.3 $\pm$ 0.52	1.000	0.420
Kōura BPUE	7.95 $\pm$ 12.15	2.8 $\pm$ 3.84	36.69 $\pm$ 36.82	0.789	0.206

Variable	Stone_pile	Control	Reference	p_vs_Con- trol	p_vs_Ref- erence
Catfish BPUE	0.15 ± 0.34	24.6 ± 55.01	97.28 ± 232.15	1.000	0.343
Substrate index (before)	3.39 ± 1.07	3.68 ± 0.95	5.29 ± 1.05	0.789	0.012
Substrate index (after)	3.65 ± 1.12	3.68 ± 0.95	5.29 ± 1.05	1.000	0.032
Slope (0–20 m)	0.06 ± 0.05	0.06 ± 0.04	0.09 ± 0.05	0.590	0.245
Temperature (°C)	24.74 ± 1.92	24.78 ± 1.47	24.48 ± 0.78	1.000	0.623
Dissolved oxygen (mg/L)	9.85 ± 1.48	9.63 ± 1.29	10.03 ± 0.86	0.418	0.426
Specific conductivity (µS/cm)	195.44 ± 70.05	199.8 ± 78.52	165.45 ± 8.5	0.281	0.951
pH	7.5 ± 2.15	7.61 ± 2.2	8.53 ± 0.66	1.000	0.426

⁸ Source: [Article Notebook](#)

⁹ Figures

¹⁰ Supplementary information

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	re- sponse	com- pari- son	time	frame	es- ti- mate	std.er- ror	statis- tic	p.value	adjco- eff	BLow	hig		
M1a	RQ1	Cat- fish	abun- dance	treat- ment	be- fore	(Intercept)	-	0.908	-	0.011	NA	-	-
							2.303		2.535		4.083	0.522	
M1a	RQ1	Cat- fish	abun- dance	treat- ment	be- fore	Site_type_de- tail	-	1.535	-	0.652	0.783	-	2.315
						Stone_pile	0.693		0.452		3.701		

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1	Q2	species	response	com- pari- type	time	frame	es- ti- mate	std.er-	stat-	p.value	adjco-	BFlow	low	high
M1ab	RQ1	Cat-	biomas	treat-	be-	(Intercept)		3.203	1.279	2.504	0.012	NA	0.696	5.709	
		fish		ment	fore										
M1ab	RQ1	Cat-	biomas	treat-	be-	Site_type_de-		-	3.107	-	0.101	0.403	-	0.990	
		fish		ment	fore	tailStone_pile		5.100		1.641			11.190		
M1aa	RQ1	Kōura	abun-	treat-	be-	(Intercept)		-	0.785	-	0.016	NA	-	-	
		dance		ment	fore			1.897		2.415			3.437	0.358	
M1aa	RQ1	Kōura	abun-	treat-	be-	Site_type_de-		0.288	1.044	0.276	0.783	0.783	-	2.334	
		dance		ment	fore	tailStone_pile							1.759		
M1ab	RQ1	Kōura	biomas	treat-	be-	(Intercept)		1.030	0.830	1.241	0.215	NA	-	2.656	
				ment	fore								0.597		
M1ab	RQ1	Kōura	biomas	treat-	be-	Site_type_de-		1.044	0.977	1.068	0.286	0.571	-	2.959	
				ment	fore	tailStone_pile							0.872		
M1ba	RQ1	Cat-	abun-	ref-	be-	(Intercept)		-	32.656	-	0.002	NA	-	-	
		fish	dance	er-	fore			102.299		3.133			166.308	38.295	
				ence											
M1ba	RQ1	Cat-	abun-	ref-	be-	Site_type_de-		-	0.536	-	0.002	0.004	-	-	
		fish	dance	er-	fore	tailReference_2		1.655		3.087			2.706	0.604	
				ence											
M1ba	RQ1	Cat-	abun-	ref-	be-	Temperature		4.079	1.293	3.154	0.002	NA	1.544	6.613	
		fish	dance	er-	fore										
				ence											
M1bb	RQ1	Cat-	biomas	ref-	be-	(Intercept)		-	37.363	-	0.000	NA	-	-	
		fish		er-	fore			153.136		4.099			226.367	79.906	
				ence											
M1bb	RQ1	Cat-	biomas	ref-	be-	Site_type_de-		-	0.743	-	0.001	0.002	-	-	
		fish		er-	fore	tailReference_2		2.569		3.457			4.026	1.112	
				ence											

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q1 species	re- response	com- pari- type	time_frame	es- ti- mate	std.er-	stat-	p.value	adjco-	BFlow	low	high
M1bb	RQ1Cat- fish	biomass	ref- er- ence	be- fore	Temperature	6.315	1.482	4.261	0.000	NA	3.410	9.220
M1ba	RQ1Kōura	abun-	ref- er- ence	be- fore	(Intercept)	0.742	0.553	1.343	0.179	NA	-	1.825
		dance									0.341	
M1ba	RQ1Kōura	abun-	ref- er- ence	be- fore	Site_type_de-	0.031	0.778	0.040	0.968	0.968	-	1.557
		dance			tailReference_2						1.494	
M1bb	RQ1Kōura	biomass	ref- er- ence	be- fore	(Intercept)	3.623	0.478	7.580	0.000	NA	2.686	4.560
M1bb	RQ1Kōura	biomass	ref- er- ence	be- fore	Site_type_de-	-	0.681	-	0.952	0.968	-	1.293
		dance			tailReference_2	0.041		0.061			1.375	
M2aa	RQ2Cat- fish	abun-	vs_confull	(Intercept)		-	1.674	-	0.028	NA	-	-
		dance	trol			3.679		2.198			6.959	0.398
M2aa	RQ2Cat- fish	abun-	vs_confull	PeriodAfter		0.184	1.470	0.125	0.900	NA	-	3.066
		dance	trol								2.698	
M2aa	RQ2Cat- fish	abun-	vs_confull	siteStone_pile		-	1.641	-	0.672	NA	-	2.522
		dance	trol			0.695		0.423			3.912	
M2aa	RQ2Cat- fish	abun-	vs_confull	monitoring_int		-	0.355	-	0.841	NA	-	0.625
		dance	trol			0.071		0.201			0.768	
M2aa	RQ2Cat- fish	abun-	vs_confull	Peri-		1.742	1.828	0.953	0.341	0.732	-	5.325
		dance	trol	odAfter:site-							1.842	
				Stone_pile								
M2ab	RQ2Cat- fish	biomass	vs_confull	(Intercept)		1.694	2.773	0.611	0.541	NA	-	7.129
		trol									3.741	

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model	Q	species	re- response	com- pari- type	time_frame	es- ti- mate	std.er-	stat-	p.value	adjco-	BFlow	low	high
M2ab	RQ26	Cat-	biomass	vs_	confull	PeriodAfter	- 2.260	- 0.709	NA	- 3.587			
		fish		trol		0.843	0.373			5.273			
M2ab	RQ26	Cat-	biomass	vs_	confull	siteStone_pile	- 5.456	- 0.341	NA	- 5.500			
		fish		trol		5.195	0.952			15.889			
M2ab	RQ26	Cat-	biomass	vs_	confull	monitoring_int	- 0.545	- 0.880	NA	- 0.985			
		fish		trol		0.082	0.151			1.150			
M2ab	RQ26	Cat-	biomass	vs_	confull	Peri-	4.977	5.352	0.930	0.352	0.732	- 15.468	
		fish		trol		odAfter:site-				5.513			
						Stone_pile							
M2ao	RQ26	Cat-	occu-	vs_	confull	(Intercept)	- 1.963	- 0.281	NA	- 1.730			
		fish	pancy	trol		2.117	1.078			5.964			
M2ao	RQ26	Cat-	occu-	vs_	confull	PeriodAfter	- 1.893	- 0.571	NA	- 2.638			
		fish	pancy	trol		1.073	0.567			4.784			
M2ao	RQ26	Cat-	occu-	vs_	confull	siteStone_pile	0.000	2.010	0.000	1.000	NA	- 3.939	
		fish	pancy	trol						3.939			
M2ao	RQ26	Cat-	occu-	vs_	confull	monitoring_int	0.213	0.331	0.644	0.519	NA	- 0.863	
		fish	pancy	trol						0.436			
M2ao	RQ26	Cat-	occu-	vs_	confull	Peri-	0.430	2.216	0.194	0.846	0.846	- 4.774	
		fish	pancy	trol		odAfter:site-				3.914			
						Stone_pile							
M2aa	RQ26	Kōura	abun-	vs_	confull	(Intercept)	- 1.178	- 0.020	NA	-	-		
		dance	trol			2.739	2.324			5.048	0.429		
M2aa	RQ26	Kōura	abun-	vs_	confull	PeriodAfter	2.121	0.973	2.179	0.029	NA	0.213	4.028
		dance	trol										
M2aa	RQ26	Kōura	abun-	vs_	confull	siteStone_pile	0.714	1.316	0.543	0.587	NA	- 3.293	
		dance	trol							1.865			
M2aa	RQ26	Kōura	abun-	vs_	confull	monitoring_int	- 0.093	- 0.189	NA	- 0.061			
		dance	trol			0.123	1.312			0.306			

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model	Q	species	re- response	com- pari- type	time_frame	es- ti- mate	std.er-	stat-	p.value	adjco-	BFlow	low	high
M2ba	RQ26	Cat-	abun-	vs_conaf-	siteStone_pile	-	2.471	-	0.274	NA	-	2.142	
		fish	dance	trol	ter	2.702		1.093			7.546		
M2ba	RQ26	Cat-	abun-	vs_conaf-	monitoring_int	-	0.474	-	0.218	NA	-	0.345	
		fish	dance	trol	ter	0.585		1.232			1.515		
M2ba	RQ26	Cat-	abun-	vs_conaf-	site-	1.171	0.611	1.917	0.055	0.121	-	2.368	
		fish	dance	trol	ter						0.026		
					Stone_pile:mon-								
					itoring_int								
M2bb	RQ26	Cat-	biomass	vs_conaf-	(Intercept)	3.626	1.976	1.835	0.066	NA	-	7.498	
		fish		trol	ter						0.247		
M2bb	RQ26	Cat-	biomass	vs_conaf-	siteStone_pile	-	4.616	-	0.162	NA	-	2.600	
		fish		trol	ter	6.448		1.397			15.496		
M2bb	RQ26	Cat-	biomass	vs_conaf-	monitoring_int	-	0.681	-	0.207	NA	-	0.474	
		fish		trol	ter	0.860		1.263			2.194		
M2bb	RQ26	Cat-	biomass	vs_conaf-	site-	1.925	1.211	1.589	0.112	0.121	-	4.299	
		fish		trol	ter						0.449		
					Stone_pile:mon-								
					itoring_int								
M2bo	RQ26	Cat-	occu-	vs_conaf-	(Intercept)	-	1.837	-	0.551	NA	-	2.505	
		fish	pancy	trol	ter	1.095		0.596			4.695		
M2bo	RQ26	Cat-	occu-	vs_conaf-	siteStone_pile	-	2.879	-	0.290	NA	-	2.594	
		fish	pancy	trol	ter	3.049		1.059			8.692		
M2bo	RQ26	Cat-	occu-	vs_conaf-	monitoring_int	-	0.487	-	0.389	NA	-	0.534	
		fish	pancy	trol	ter	0.419		0.862			1.373		
M2bo	RQ26	Cat-	occu-	vs_conaf-	site-	1.128	0.701	1.610	0.107	0.121	-	2.502	
		fish	pancy	trol	ter						0.245		
					Stone_pile:mon-								
					itoring_int								
M2ba	RQ26	Kōura	abun-	vs_conaf-	(Intercept)	-	0.911	-	0.886	NA	-	1.655	
			dance	trol	ter	0.130		0.143			1.915		

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model	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50	Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60	Q61	Q62	Q63	Q64	Q65	Q66	Q67	Q68	Q69	Q70	Q71	Q72	Q73	Q74	Q75	Q76	Q77	Q78	Q79	Q80	Q81	Q82	Q83	Q84	Q85	Q86	Q87	Q88	Q89	Q90	Q91	Q92	Q93	Q94	Q95	Q96	Q97	Q98	Q99	Q100
re-	com-	pari-	time	frame	es-	ti-	std.er-	stat-	p.value	adjco	BFlow	low	high																																																																																							
M2ba	RQ2K	ōura	abun-	vs_conaf-	site	Stone_pile	0.177	1.231	0.144	0.886	NA	-	2.589																																																																																							
		dance	trol	ter								2.235																																																																																								
M2ba	RQ2K	ōura	abun-	vs_conaf-	monitoring_int		-	0.164	-	0.044	NA	-	-																																																																																							
		dance	trol	ter			0.329		2.014			0.650	0.009																																																																																							
M2ba	RQ2K	ōura	abun-	vs_conaf-	site-		0.318	0.205	1.552	0.121	0.121	-	0.718																																																																																							
		dance	trol	ter	Stone_pile:mon-							0.083																																																																																								
					itoring_int																																																																																															
M2bb	RQ2K	ōura	biomass	vs_conaf-	(Intercept)		3.105	0.891	3.487	0.000	NA	1.360	4.850																																																																																							
			trol	ter																																																																																																
M2bb	RQ2K	ōura	biomass	vs_conaf-	site	Stone_pile	-	1.190	-	0.821	NA	-	2.062																																																																																							
			trol	ter			0.270		0.227			2.601																																																																																								
M2bb	RQ2K	ōura	biomass	vs_conaf-	monitoring_int		-	0.220	-	0.124	NA	-	0.093																																																																																							
			trol	ter			0.339		1.538			0.771																																																																																								
M2bb	RQ2K	ōura	biomass	vs_conaf-	site-		0.437	0.276	1.584	0.113	0.121	-	0.977																																																																																							
			trol	ter	Stone_pile:mon-							0.104																																																																																								
					itoring_int																																																																																															
M2bo	RQ2K	ōura	æccu-	vs_conaf-	(Intercept)		1.529	2.376	0.643	0.520	NA	-	6.185																																																																																							
		pancy	trol	ter								3.128																																																																																								
M2bo	RQ2K	ōura	æccu-	vs_conaf-	site	Stone_pile	-	3.580	-	0.686	NA	-	5.571																																																																																							
		pancy	trol	ter			1.447		0.404			8.465																																																																																								
M2bo	RQ2K	ōura	æccu-	vs_conaf-	monitoring_int		-	0.398	-	0.287	NA	-	0.356																																																																																							
		pancy	trol	ter			0.424		1.066			1.203																																																																																								
M2bo	RQ2K	ōura	æccu-	vs_conaf-	site-		1.325	0.827	1.602	0.109	0.121	-	2.946																																																																																							
		pancy	trol	ter	Stone_pile:mon-							0.296																																																																																								
					itoring_int																																																																																															
M3aa	RQ3C	at-	abun-	vs_ref-af-	(Intercept)		-	1.347	-	0.000	NA	-	-																																																																																							
		fish	dance	er-			6.558		4.869			9.198	3.918																																																																																							
				ence																																																																																																

Table 2: Full fixed-effect estimates (, SE, 95% CI, p-value) for every term in every model currently in model_registry. Unlike the interaction-only summary above, this surfaces covariate effects (e.g. Substrate_index_after, Season) directly rather than requiring a one-off check per model.

model	Q	species	re- response	com- pari- type	time_frame	es- ti- mate	std.er- ror	stat- tic	p.value	adjco- B	Flow	low	high
M3aadRQ36	Cat	abun-	vs_ref-	af-	siteStone_pile	-	2.048	-	0.185	NA	-	1.298	
	fish	dance	er-	ter		2.717		1.326			6.732		
			ence										
M3aadRQ36	Cat	abun-	vs_ref-	af-	monitoring_int	-	0.158	-	0.364	NA	-	0.166	
	fish	dance	er-	ter		0.143		0.907			0.453		
			ence										
M3aadRQ36	Cat	abun-	vs_ref-	af-	Temperature	0.273	0.065	4.183	0.000	NA	0.145	0.401	
	fish	dance	er-	ter									
			ence										
M3aadRQ36	Cat	abun-	vs_ref-	af-	site-	0.579	0.434	1.334	0.182	0.243	-	1.429	
	fish	dance	er-	ter	Stone_pile:mon-						0.271		
			ence		itoring_int								
M3abRQ36	Cat	biomass	vs_ref-	af-	(Intercept)	2.334	1.213	1.924	0.054	NA	-	4.711	
	fish		er-	ter							0.044		
			ence										
M3abRQ36	Cat	biomass	vs_ref-	af-	siteStone_pile	-	4.117	-	0.102	NA	-	1.333	
	fish		er-	ter		6.736		1.636			14.806		
			ence										
M3abRQ36	Cat	biomass	vs_ref-	af-	monitoring_int	-	0.222	-	0.919	NA	-	0.413	
	fish		er-	ter		0.023		0.102			0.458		
			ence										
M3abRQ36	Cat	biomass	vs_ref-	af-	site-	1.300	0.923	1.407	0.159	0.243	-	3.109	
	fish		er-	ter	Stone_pile:mon-						0.510		
			ence		itoring_int								
M3aadRQ36	Kōura	abun-	vs_ref-	af-	(Intercept)	0.307	0.533	0.575	0.565	NA	-	1.352	
		dance	er-	ter							0.739		
			ence										

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model	Q	species	response	com- pari- type	time_frame	es- ti- mate	std.er-	stat- tic	p.value	adjco-	BFlow	low	high
M3aa	RQ3K	Kōura	abun-	vs_ref-af-	siteStone_pile	-	0.923	-	0.805	NA	-	1.582	
			dance	er-ter		0.228		0.247			2.038		
				ence									
M3aa	RQ3K	Kōura	abun-	vs_ref-af-	monitoring_int	-	0.072	-	0.101	NA	-	0.023	
			dance	er-ter		0.118		1.640			0.260		
				ence									
M3aa	RQ3K	Kōura	abun-	vs_ref-af-	site-	0.100	0.131	0.760	0.447	0.447	-	0.358	
			dance	er-ter	Stone_pile:mon-						0.158		
				ence	itoring_int								
M3ab	RQ3K	Kōura	biomass	vs_ref-af-	(Intercept)	3.445	0.539	6.393	0.000	NA	2.389	4.502	
				er-ter									
				ence									
M3ab	RQ3K	Kōura	biomass	vs_ref-af-	siteStone_pile	-	0.934	-	0.488	NA	-	1.182	
				er-ter		0.648		0.694			2.477		
				ence									
M3ab	RQ3K	Kōura	biomass	vs_ref-af-	monitoring_int	-	0.097	-	0.178	NA	-	0.060	
				er-ter		0.131		1.346			0.322		
				ence									
M3ab	RQ3K	Kōura	biomass	vs_ref-af-	site-	0.231	0.173	1.338	0.181	0.243	-	0.569	
				er-ter	Stone_pile:mon-						0.107		
				ence	itoring_int								
M3ba	RQ3K	Cat-	abun-	ref-af-	(Intercept)	-	1.788	-	0.000	NA	-	-	
		fish	dance	er-ter		6.715		3.757			10.219	13.212	
				ence									
M3ba	RQ3K	Cat-	abun-	ref-af-	Site_type_de-	0.731	1.033	0.707	0.479	NA	-	2.756	
		fish	dance	er-ter	tailReference_2						1.295		
				ence									

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model	Q	species	re- response	com- pari- type	time_frame	es- ti- mate	std.er-	stat- tic	p.value	adjco-	BFlow	low	high
M3ba	RQ36	Cat- fish	abun- dance	ref- er- ence	af- ter	monitoring_int	- 0.051	0.233 0.220	- 0.826	NA 0.509	- 0.406		
M3ba	RQ36	Cat- fish	abun- dance	ref- er- ence	af- ter	Temperature	0.258	0.082	3.131	0.002	NA	0.096	0.419
M3ba	RQ36	Cat- fish	abun- dance	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2:moni- toring_int (Intercept)	- 0.146	0.296 0.491	- 0.623	0.902 0.727	- 0.435		
M3bb	RQ36	Cat- fish	biomass	ref- er- ence	af- ter	(Intercept)	2.128	1.527	1.393	0.163	NA	- 0.865	5.121
M3bb	RQ36	Cat- fish	biomass	ref- er- ence	af- ter	Site_type_de- tailReference_2	0.062	1.575	0.039	0.969	NA	- 3.025	3.149
M3bb	RQ36	Cat- fish	biomass	ref- er- ence	af- ter	monitoring_int	0.007	0.325	0.023	0.982	NA	- 0.629	0.644
M3bb	RQ36	Cat- fish	biomass	ref- er- ence	af- ter	Site_type_de- tailRefer- ence_2:moni- toring_int	0.055	0.448	0.123	0.902	0.902	- 0.823	0.933
M3ba	RQ36	Kōura	abun- dance	ref- er- ence	af- ter	(Intercept)	0.712	0.805	0.884	0.377	NA	- 0.866	2.290

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model	Q	species	response	com- pari- type	time_frame	es- ti- mate	std.er-	stat- tic	p.value	adjco-	BFlow	low	high
M3ba	RQ3K	Kōura	abun-	ref-	af-	Site_type_de-	-	0.567	-	0.642	NA	-	0.848
			dance	er-	ter	tailReference_2	0.263	0.464			1.374		
				ence									
M3ba	RQ3K	Kōura	abun-	ref-	af-	monitoring_int	-	0.075	-	0.069	NA	-	0.010
			dance	er-	ter		0.136	1.820			0.283		
				ence									
M3ba	RQ3K	Kōura	abun-	ref-	af-	SeasonSpring	-	0.210	-	0.103	NA	-	0.070
			dance	er-	ter		0.343	1.628			0.755		
				ence									
M3ba	RQ3K	Kōura	abun-	ref-	af-	SeasonSummer	-	0.225	-	0.349	NA	-	0.231
			dance	er-	ter		0.211	0.936			0.652		
				ence									
M3ba	RQ3K	Kōura	abun-	ref-	af-	SeasonWinter	-	0.242	-	0.000	NA	-	-
			dance	er-	ter		0.942	3.889			1.417	0.467	
				ence									
M3ba	RQ3K	Kōura	abun-	ref-	af-	Site_type_de-	-	0.101	-	0.802	0.902	-	0.173
			dance	er-	ter	tailRefer-	0.025	0.251			0.223		
				ence		ence_2:moni-							
						toring_int							
M3bb	RQ3K	Kōura	biomass	ref-	af-	(Intercept)	3.370	0.860	3.917	0.000	NA	1.683	5.056
				er-	ter								
				ence									
M3bb	RQ3K	Kōura	biomass	ref-	af-	Site_type_de-	-	0.737	-	0.803	NA	-	1.261
				er-	ter	tailReference_2	0.184	0.250			1.630		
				ence									
M3bb	RQ3K	Kōura	biomass	ref-	af-	monitoring_int	-	0.117	-	0.373	NA	-	0.125
				er-	ter		0.104	0.891			0.334		
				ence									

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model	Q1	Q3	Kourbiomassref- er- ence	time_frame	Site_type_de- tailRefer- ence_2:moni- toring_int	es- ti- mate	std.er- ror	stat- tic	p.value	adjco- BFlow	low	high
M3bb	RQ3K			af- ter		-	0.167	-	0.812	0.902	-	0.288
						0.040		0.237			0.368	

¹¹ Source: [Article Notebook](#)

Table 3: DHARMa goodness-of-fit tests per model. $p < 0.05$ indicates a significant departure from model assumptions (uniformity of residuals, over/under-dispersion, or excess outliers) and warrants closer inspection of the corresponding diagnostic plot above.

Model	Uniformity_p	Dispersion_p	Outlier_p
M2aac	0.139	0.952	0.11
M2abc	0.004	0.482	1.00
M2aoc	0.289	0.770	1.00
M2aak	0.708	0.906	1.00
M2abk	0.886	0.816	1.00
M2aok	0.331	0.952	1.00
M2bac	0.024	0.776	1.00
M2bbc	0.109	0.596	1.00
M2boc	0.084	0.638	1.00
M2bak	0.796	0.740	1.00
M2bbk	0.505	0.704	1.00
M2bok	0.468	0.896	1.00
M3aac	0.860	0.606	1.00
M3abc	0.551	0.346	1.00
M3aak	0.618	0.910	1.00
M3abk	0.994	0.724	1.00

Table 3: DHARMA goodness-of-fit tests per model. $p < 0.05$ indicates a significant departure from model assumptions (uniformity of residuals, over/under-dispersion, or excess outliers) and warrants closer inspection of the corresponding diagnostic plot above.

Model	Uniformity_p	Dispersion_p	Outlier_p
M3bbc	0.885	0.748	1.00
M3bak	0.959	0.878	1.00
M3bbk	0.872	0.896	1.00

12 Source: [Article Notebook](#)

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Figure 1: Map of the monitoring sites.

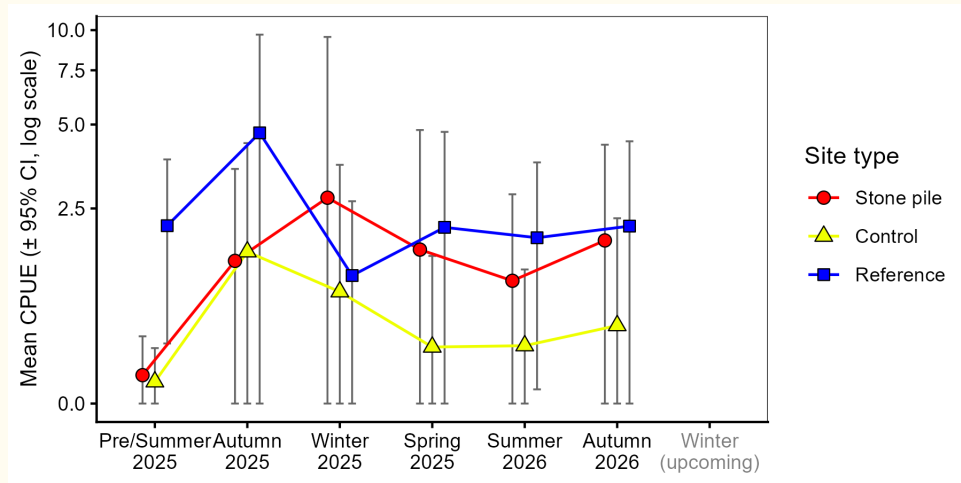


Figure 2: Mean catch per monitoring round ($\pm$ 95% CI) by site type for kōura.

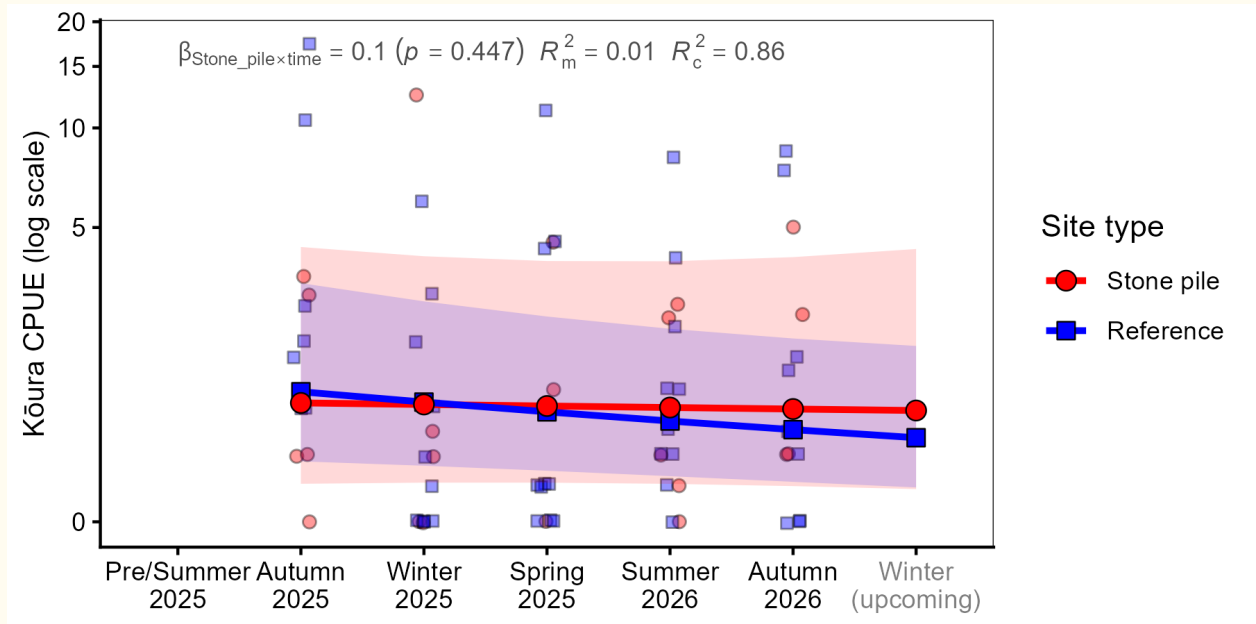


Figure 3: Predicted kōura CPUE trajectories over monitoring rounds at stone pile and reference sites from the RQ3 abundance convergence model. Shaded ribbons represent 95% confidence intervals. The annotation shows the key interaction term ($\pm$ p-value) and model R^2 values for fixed effects (R^2_m) and conditional (R^2_c).

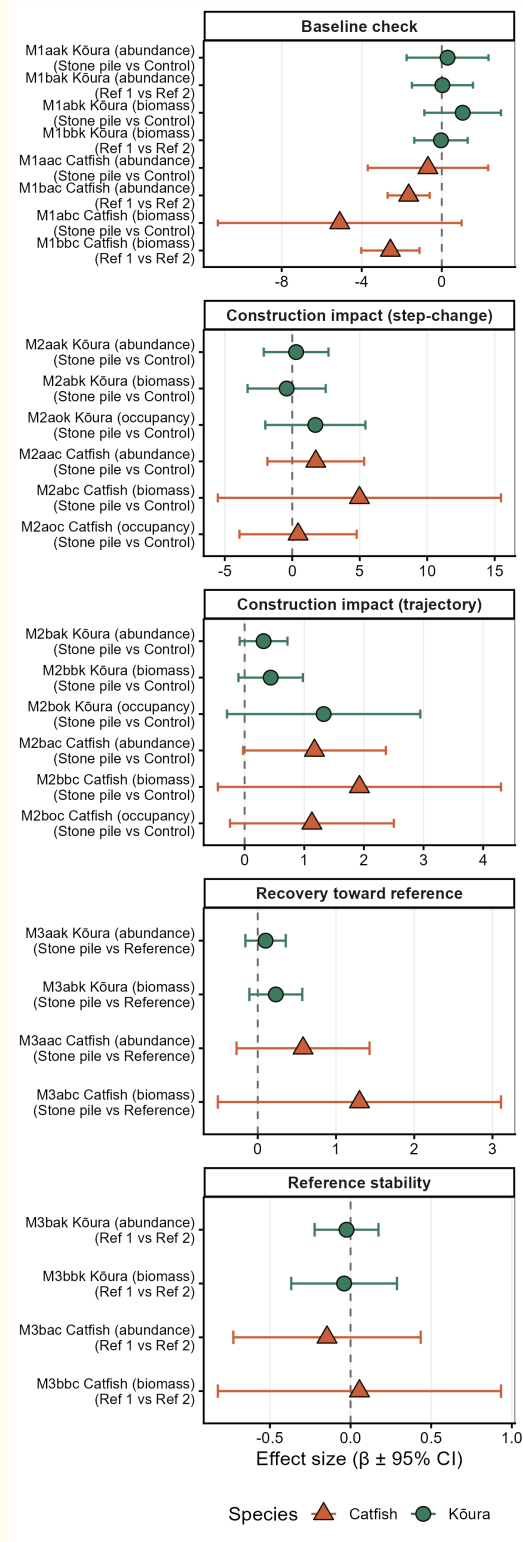


Figure 4: Effect sizes ($\beta \pm 95\%$ CI) for the key term of interest of every model in model_registry (kōura and catfish, RQ1–RQ4). Filled points indicate significant effects ($p < 0.05$). Each panel has its own x-axis scale— magnitudes are not directly comparable across panels because each RQ tests a different contrast (step-change, per-round slope, or simple group difference) and RQ1 occupancy is on the log-odds scale while all others use the Tweedie log link. Models with degenerate confidence intervals (non-positive-definite Hessian) are excluded.

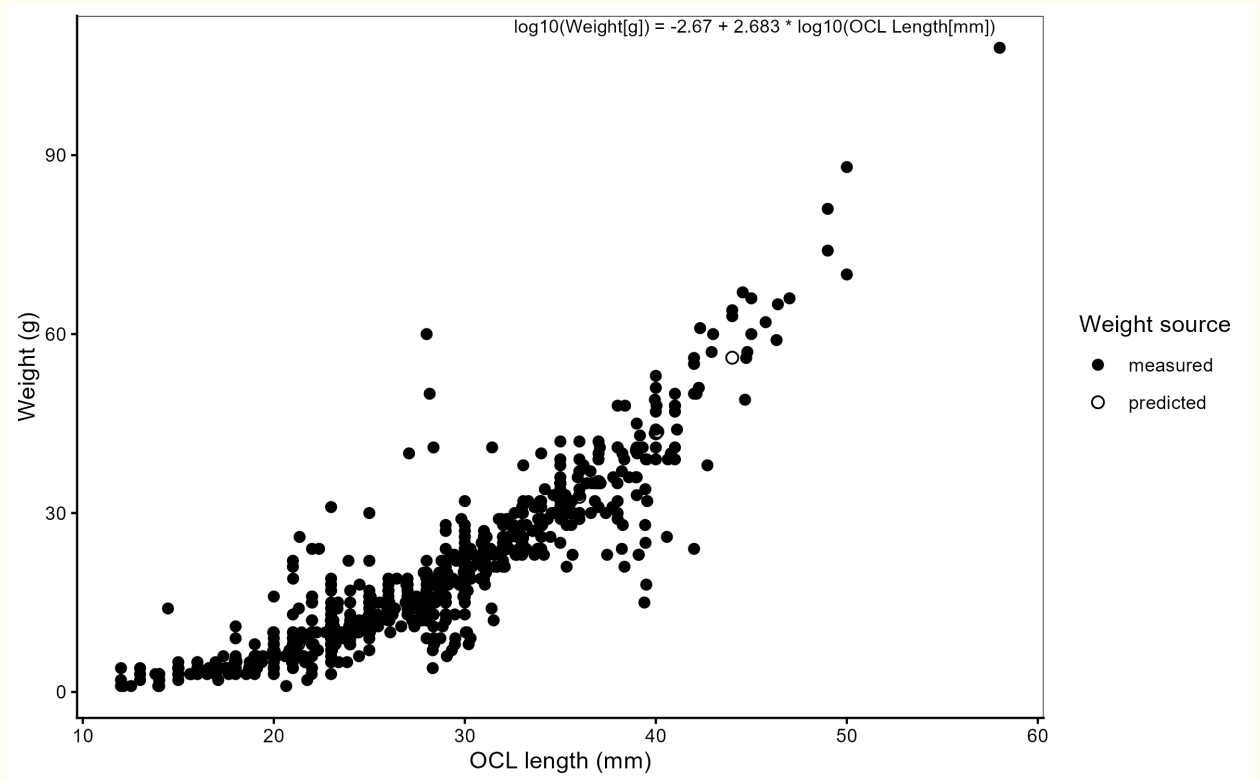


Figure 5: Length–weight relationship for kōura, fitted on log –log scale. Open points are weights predicted from the length–weight regression (used to fill missing scale measurements); filled points are directly measured weights.

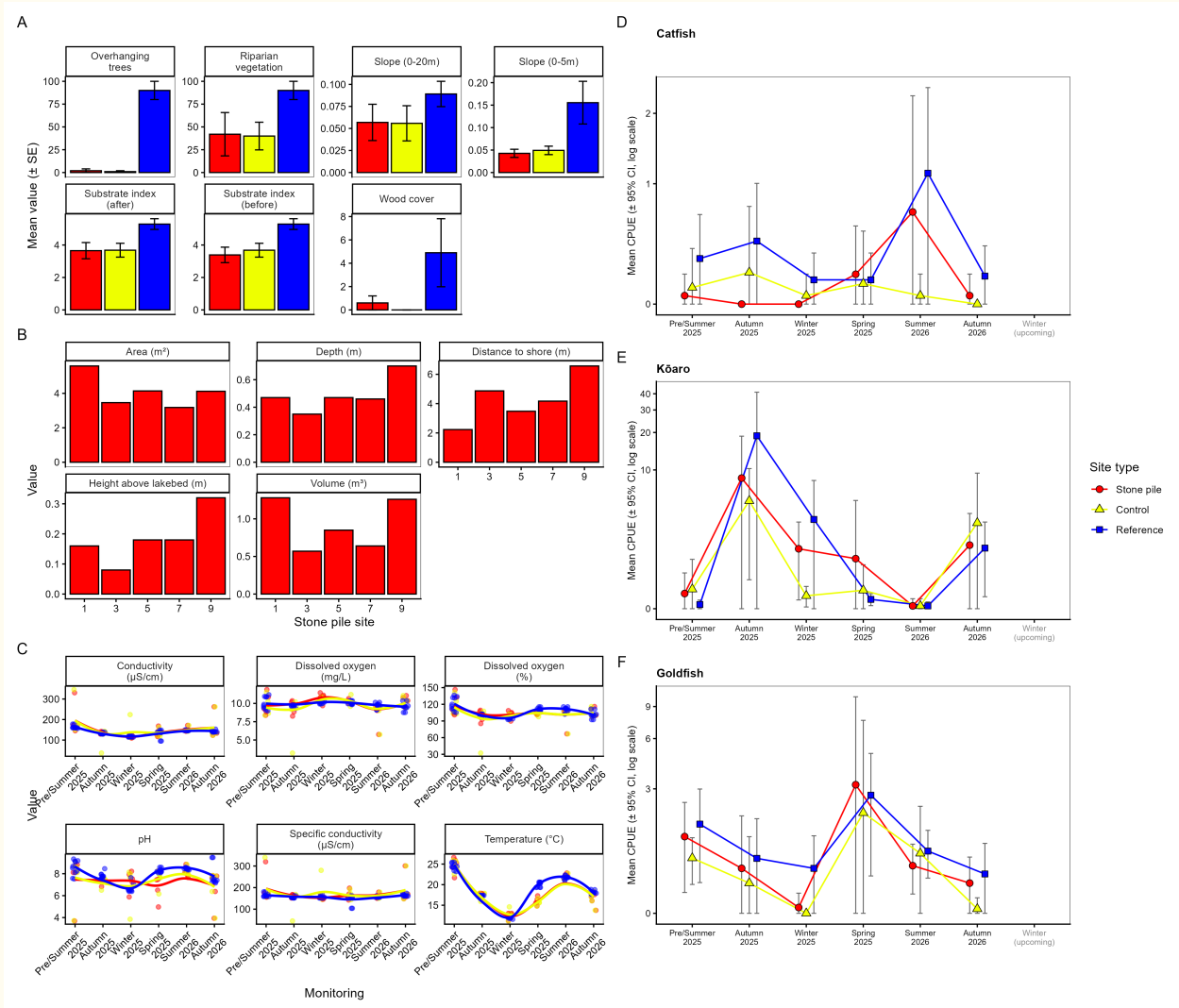


Figure 6: Descriptive context plots, combined. Left column (top to bottom): baseline habitat characteristics (mean $\pm$ SE by site type), stone pile structure dimensions (5 stone pile sites), chemical conditions over time (mean $\pm$ smoothed trend by site type). Right column: mean catch per monitoring round by site type, for catfish, kōaro, and goldfish. Site-type colours are shared across all panels — see the legend on the right. Background/methods-supporting figures, not primary results — see Results for the BACI, trend, and convergence models.

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